Figure Descriptions- Sequential Time-Warping Hidden Markov Model

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Generative Model: Simulated Data

The top row of heatmaps shows the model parameters which were hardcoded for simulating data and serve as ground-truth for testing modeling fitting. Generated by the sequential structure of the *Transition Matrix* are two sequences, with both forward and reverse realizations by the clone structure of the *Emissions Matrix*. The dynamics of the time-warp variable are given by the random walk with jumps in the *Time-Warp Matrix*. The bottom plot shows the simulated time-warp, latent (hidden) state trajectory and the spike counts.

Model Fitting: Expectation—Maximization

The plot on the left shows the complete data log-likelihood before each step of the EM algorithm for parameter estimation (see methods). 100 random initializations were used to fit 200 seconds of simulated data, and the distribution of the final log-likelihood values are shown in the histogram on the right. The best model was that which had the minimum negative log-likelihood (maximum likelihood).

Model Fitting: Recovered Parameters

The top row of heatmaps is the same as in the generative model slide—these are the ground truth parameters which simulated the data. The bottom row is the recovered parameters. Notice that the recovered transition matrix and emissions matrix are nearly identical to the ground truth parameters. Note that there is no clear mapping between inferred and ground-truth latent states. However, we can clearly see that while the first and second sequences are flipped, the structure is otherwise identical. This implies the model correctly learned the emissions. Fitting the time-warp matrix needs some additional work. One possible reason for the difficulty in fitting the time-warp matrix is that there are no constraints on this matrix, whereas the other two matrixes are constrained with sequences and clones (see methods).

Model Fitting: Tau Smoothing

Since the Time-Warp matrix fit poorly I wanted to know if the inference of the time-warp variable worked at all. I found that by smoothing the inferred time-warp variable we can recover the ground-truth time-warp well. The plot on the left shows the raw, smoothed, and ground truth time-warp with the correlation and RMSE between the inferred and ground-truth values. This result makes me think we should constrain the time-warp matrix during fitting. (but I am not sure what assumptions to make about time-warp dynamics which are biologically plausible. Does the random walk with jumps which I used to generate the data seem reasonable?)

Model Inference: Viterbi and Neuron Sorting

After learning the model parameters, inference is run to recover the latent state and time-warp trajectories from the spike counts using the modified Viterbi algorithm (see methods). These two trajectories are plotted in the top two rows. Please note that due to the lack of clear mapping between inferred and ground-truth latent states discussed above, these trajectories at first look different. However, a close inspection reveals that the model correctly predicts the beginning and end of each sequence and internally categorizes sequences consistently. Also, the inferred and smoothed time-warp tracks well with the ground truth. Finally, the neurons in the spike raster are sorted based on their inferred tuning to latent states (sorting the rows of the emissions matrix). We can observe that learned neuron sorting recovers the exact same sequences as in the simulated data.